
1. PROJECT INTRODUCTION

1.1 Project Overview

In today's digital world, web applications are widely used in banking, e-commerce, healthcare, education, and various business sectors. As the usage of web applications increases, security threats against these applications also continue to grow. Attackers often exploit vulnerabilities such as SQL Injection (SQLi), Cross-Site Scripting (XSS), Broken Authentication, and Security Misconfigurations to gain unauthorized access to sensitive information.

To ensure that web applications are secure and resistant to cyberattacks, security testing has become an essential part of the Software Testing Life Cycle (STLC). Manual security testing can be time-consuming, repetitive, and prone to human errors. Therefore, organizations are increasingly adopting automated security testing solutions.

This project, "**Web Application Security Testing of DVWA using OWASP ZAP**", aims to automate the process of identifying security vulnerabilities within a web application. The project uses **Damn Vulnerable Web Application (DVWA)** as the target application because it contains intentionally vulnerable modules designed for security testing and learning purposes.

The automation framework is developed using **Java, Selenium WebDriver, TestNG, Maven, and OWASP ZAP**. Selenium WebDriver is used to automate user actions such as login and navigation. SQL Injection and Cross-Site Scripting (XSS) vulnerabilities are tested using automated test cases. OWASP ZAP is integrated into the framework to perform automated Spider Scans and Active Scans for vulnerability identification.

The project also generates TestNG Reports, Extent Reports, and OWASP ZAP Security Reports, providing detailed information about test execution and detected vulnerabilities. The overall goal is to demonstrate how automation can improve the effectiveness, consistency, and efficiency of web application security testing.

1.2 Project Objective

The primary objective of this project is to develop an automated security testing framework capable of identifying security vulnerabilities within a web application and generating comprehensive security reports.

The specific objectives of the project are as follows:

- To automate the login functionality of DVWA using Selenium WebDriver.
- To verify whether the application properly handles valid and malicious user inputs.
- To perform automated SQL Injection testing using different SQL payloads.

- To perform automated Cross-Site Scripting (XSS) testing using multiple JavaScript payloads.
- To integrate OWASP ZAP with the automation framework for vulnerability scanning.
- To execute Spider Scans and Active Scans using OWASP ZAP.
- To identify security vulnerabilities and categorize them based on risk severity.
- To generate TestNG and Extent Reports for test execution analysis.
- To generate OWASP ZAP Security Reports containing vulnerability details.
- To provide security recommendations and mitigation strategies for identified vulnerabilities.
- To demonstrate the practical implementation of Security Testing Automation in a controlled environment.

The project serves as a learning platform for understanding web application security testing, automation framework development, and vulnerability assessment techniques.

1.3 Problem Statement

Web applications frequently accept user input through login forms, search fields, registration pages, and URL parameters. If user inputs are not properly validated and sanitized, attackers can exploit vulnerabilities to gain unauthorized access, manipulate data, or execute malicious scripts.

Many organizations still depend on manual security testing approaches. Although manual testing can identify vulnerabilities, it often presents several challenges:

- Testing activities require significant time and effort.
- Repetitive security tests are difficult to perform consistently.
- Human errors may cause vulnerabilities to remain undetected.
- Security validation becomes difficult when applications are updated frequently.
- Large-scale applications require substantial resources for complete security assessment.

Among the most common web application vulnerabilities are SQL Injection and Cross-Site Scripting (XSS). These vulnerabilities continue to appear in many applications despite the availability of security standards and secure coding practices.

The challenge addressed by this project is to automate the security testing process so that vulnerabilities can be identified quickly, accurately, and consistently. By integrating Selenium automation with OWASP ZAP, the project provides a practical solution for performing repeatable security testing and generating detailed vulnerability reports.

This approach helps improve software quality, enhance security posture, and support secure software development practices.

1.4 Scope of Project

The scope of this project focuses on automating security testing activities for the Damn Vulnerable Web Application (DVWA) in a controlled testing environment.

The project covers the following areas:

Login Testing

The login functionality of DVWA is automated using Selenium WebDriver. The objective is to verify successful authentication and navigation to the application's home page after login.

SQL Injection Testing

The SQL Injection module of DVWA is tested using multiple malicious SQL payloads. The testing process evaluates whether the application is vulnerable to unauthorized database access and data manipulation attacks.

Cross-Site Scripting (XSS) Testing

The XSS module is tested using various JavaScript payloads. The purpose is to determine whether user inputs are properly sanitized before being displayed within the application.

OWASP ZAP Security Scanning

OWASP ZAP is integrated with the automation framework to perform:

- Spider Scanning
- Active Scanning
- Vulnerability Detection
- Risk Classification
- Security Report Generation

Test Automation Framework

The project includes the development of an automation framework using:

- Java
- Selenium WebDriver
- TestNG
- Maven
- OWASP ZAP

Reporting

The framework generates:

- TestNG Reports
- Extent Reports
- OWASP ZAP Security Reports

Out of Scope

The project does not include:

- Production environment testing
- Real-world penetration testing
- Denial of Service (DoS) attacks
- Network security testing
- Mobile application security testing
- Cloud infrastructure security assessment

The project is strictly limited to web application security testing within a local testing environment.

1.5 Expected Outcomes

Upon successful completion of the project, several important outcomes are expected.

The automation framework will be capable of executing functional and security test cases automatically without requiring manual intervention. The framework will successfully automate login functionality, SQL Injection testing, XSS testing, and vulnerability scanning activities.

The project is expected to identify common web application vulnerabilities present within DVWA, including:

- SQL Injection
- Cross-Site Scripting (XSS)
- Missing Security Headers
- Security Misconfigurations
- Information Disclosure Issues

OWASP ZAP integration will enable automated vulnerability scanning and provide detailed information about detected security risks. The generated security reports will help developers and testers understand potential threats and implement corrective measures.

The project will also produce TestNG and Extent Reports containing execution summaries, pass/fail statistics, execution logs, and screenshots. These reports improve traceability and simplify result analysis.

From a learning perspective, the project will enhance knowledge and practical experience in:

- Security Testing
- Vulnerability Assessment
- Selenium Automation
- TestNG Framework
- OWASP ZAP Integration

- Security Reporting
- Secure Software Development Practices

The final outcome will be a complete Security Testing Automation Framework capable of identifying vulnerabilities, generating reports, and demonstrating industry-standard security testing techniques using modern automation tools.
